1.01 Indices

Worked Solutions

1 a)

2

1 b)

$3^{-1} = \frac{1}{3}$

1 c)

$7^2 = 49$

1 d)

$2 \times 27 \times \frac{1}{9} = 6$

1 e)

$243 \div \frac{1}{3} = 729$

1 f)

$(4 \times 2)^{-1} = \frac{1}{8}$

1 g)

$(\frac{1}{4})^{-1} = 4$

1 h)

$(\frac{3}{2})^{-1} = \frac{2}{3}$

1 i)

$\frac{4^2}{2^5} = \frac{16}{32} = \frac{1}{2}$

2 a)

$x^1 x^{\frac{1}{2}} = x^{\frac{3}{2}}$

2 b)

$x^2 x^{\frac{1}{2}} = x^{\frac{5}{2}}$

2 c)

$x^1 x^{-\frac{1}{2}} = x^{\frac{1}{2}}$

2 d)

$x^2 x^{-\frac{1}{2}} = x^{\frac{3}{2}}$

2 e)

$x^{\frac{1}{2}}x^{\frac{1}{2}}x^{\frac{1}{2}} = x^{\frac{3}{2}}$

2 f)

$x^{\frac{1}{3}}x^{\frac{1}{3}} = x^{\frac{2}{3}}$

2 g)

$x^{-\frac{1}{2}}$

2 h)

$4x^{-\frac{3}{2}}$

2 i)

$x^{1}x^{\frac{1}{2}}x^{-\frac{1}{3}} = x^{\frac{7}{6}}$

3 a)

$\sqrt{x}$

3 b)

$\frac{1}{\sqrt{x}}$

3 c)

$\frac{1}{x^5}$

3 d)

$\sqrt[3]{x^2}$

3 e)

$\frac{1}{\sqrt[4]{x^3}}$

3 f)

$x^{\frac{3}{5}} = \sqrt[5]{x^3}$

4 a)

$4^{\frac{3}{2}}a^{10\times\frac{3}{2}} = 8a^{15}$

4 b)

$a^{1}b^{\frac{2}{3}}c^{\frac{1}{3}} \times 2a^{2}b^{1}c^{\frac{1}{2}} = 2a^{3}b^{\frac{5}{3}}c^{\frac{5}{6}}$

4 c)

$\frac{4a^{\frac{1}{2}}b^{1}}{3a^{2}b} = \frac{4}{3}a^{-\frac{3}{2}}$

4 d)

$4a^6b^{-1}c^2$

4 e)

$\frac{9a^4b^6c^2}{3ab^{-1}c^2} = 3a^3b^7$

5 a)

$\frac{8a^3 \times 243a^{10}}{36a^{10}} = 54a^3$

5 b)

$\frac{27a^3 \times 16a^4}{8a^{12}} = 54a^{-5}$

5 c)

$\frac{(a^3)^4}{a^{\frac{1}{2}} \times a^{\frac{5}{2}}} = \frac{a^{12}}{a^3} = a^9$

5 d)

$\frac{8}{a^{\frac{24}{5}}} \times \frac{a^7}{4} = 2a^{\frac{11}{5}}$

6 a)

$(x^8)^{\frac{1}{2}} = x^4$

6 b)

$(x^{\frac{4}{5}})^{\frac{1}{2}} = x^{\frac{2}{5}}$

6 c)

$\frac{7x^3y^2}{3xy^3} = \frac{7x^2}{3y} = \frac{7}{3}x^2y^{-1}$

6 d)

$\frac{4x^2y^3}{(9x^2y^3)^{\frac{1}{2}}} = \frac{4x^2y^3}{3xy^{\frac{3}{2}}} = \frac{4}{3}xy^{\frac{3}{2}}$

7 a)

$x^{\frac{1}{2}}(2x+1) = 2x^{\frac{3}{2}} + x^{\frac{1}{2}}$

7 b)

$(x^{\frac{1}{2}}+1)(x^{\frac{1}{2}}-2) = x + x^{\frac{1}{2}} - 2x^{\frac{1}{2}} - 2 = x - x^{\frac{1}{2}} - 2$

7 c)

$\frac{x+1}{x^{\frac{1}{2}}} = x^{\frac{1}{2}} + x^{-\frac{1}{2}}$

7 d)

$\frac{x^2-1}{x^{\frac{1}{2}}} = x^{\frac{3}{2}} - x^{-\frac{1}{2}}$

7 e)

$4x^{-1} + \frac{1}{3}x^{-\frac{1}{2}}$

7 f)

$\frac{x^{\frac{1}{2}}-3}{x \times x^{\frac{1}{2}}} = \frac{x^{\frac{1}{2}}-3}{x^{\frac{3}{2}}} = x^{-1} - 3x^{-\frac{3}{2}}$

8 a)

$\sqrt{(7a^{\frac{1}{2}})^2 + (24a^{\frac{1}{2}})^2} = \sqrt{49a + 576a} = \sqrt{625a} = 25\sqrt{a}$

8 b)

$\frac{1}{2} \times 7a^{\frac{1}{2}} \times 24a^{\frac{1}{2}} = 84a$

9 a)

$x = 2^2 = 4$

9 b)

$\frac{1}{x} = 3^2 \Rightarrow x = \frac{1}{9}$

9 c)

$x = \frac{1}{2^3} = \frac{1}{8}$

9 d)

$\frac{1}{x} = 11^2 \Rightarrow x = \frac{1}{121}$

9 e)

$\frac{1}{x} = (\frac{8}{27})^{\frac{2}{3}} = \frac{4}{9} \Rightarrow x = \frac{9}{4}$

9 f)

$\frac{1}{x} = (\frac{3}{2})^3 = \frac{27}{8} \Rightarrow x = \frac{8}{27}$

9 g)

$\frac{1}{x} = (\frac{25}{9})^{\frac{3}{2}} = \frac{125}{27} \Rightarrow x = \frac{27}{125}$

10)

His mistake is that in finding the 7th root, he identifies -3 as an answer.

$$x^{\frac{1}{4}} = \sqrt[7]{2187} = 3$$
$$x = 3^4$$
$$x = 81$$

11 a)

$8^{\frac{1}{2}}$

11 b)

$8^{10-12} = 8^{-2}$

11 c)

$\frac{1}{8^2} = 8^{-2}$

11 d)

$64^{\frac{1}{3}} = (8^2)^{\frac{1}{3}} = 8^{\frac{2}{3}}$

11 e)

$\frac{8^{\frac{1}{3}}}{8^{\frac{1}{2}}} = 8^{-\frac{1}{6}}$

11 f)

$\frac{(8^{\frac{1}{3}})^6}{8^{\frac{1}{2}}} = 8^{\frac{3}{2}}$

11 g)

$(8^{\frac{4}{3}})^{\frac{1}{3}} \times (8^3)^{\frac{1}{9}} = 8^{\frac{4}{9}} \times 8^{\frac{1}{3}} = 8^{\frac{7}{9}}$

11 h)

$8^{\frac{1}{3}} \times 8 \times (8^{\frac{4}{3}})^{\frac{1}{3}} = 8^{\frac{1}{3}} \times 8^1 \times 8^{\frac{4}{9}} = 8^{\frac{16}{9}}$

12 a)

$x + \frac{1}{2} = 1 \Rightarrow x = \frac{1}{2}$

12 b)

$x + \frac{1}{2} = -2 \Rightarrow x = -\frac{5}{2}$

12 c)

$x + 1 = 2x \Rightarrow x = 1$

12 d)

$(2^2)^{2x} = 2^{\frac{1}{2}}$

$4x = \frac{1}{2} \Rightarrow x = \frac{1}{8}$

12 e)

$3x - 1 = 5 \Rightarrow x = 2$

12 f)

$3x + 2 = 2(x + 5) \Rightarrow 3x + 2 = 2x + 10 \Rightarrow x = 8$

12 g)

$(3^2)^{2x-1} = 3^{-\frac{1}{2}}$

$2(2x - 1) = -\frac{1}{2} \Rightarrow 4x - 2 = -\frac{1}{2} \Rightarrow x = \frac{3}{8}$

12 h)

$\frac{3^{2x}}{3^{x+2}} = 3^3 \Rightarrow 2x - (x+2) = 3 \Rightarrow x = 5$

12 i)

$3^{x^2-x} = 6^3$

$x^2 - x - 6 = 0 \Rightarrow x = -2 \text{ or } x = 3$

13 a)

$4 \times 4^x = 4^1 \times 4^x = 4^{1+x}$

13 b)

$2^{2x} + (2^2)^x = 2 \times 2^{2x} = 2^1 \times 2^{2x} = 2^{2x+1}$

13 c)

$3 \times 2^{3x} + 5 \times (2^3)^x = 8 \times 2^{3x} = 2^3 \times 2^{3x} = 2^{3x+3}$

13 d)

$5 \times (2^4)^{2x} - 3 \times 2^{8x} = 5 \times 2^{8x} - 3 \times 2^{8x} = 2^1 \times 2^{8x} = 2^{1+8x}$

13 e)

$3^1 \times 3^{2x} + 3^{2x} = 4 \times 3^{2x}$

13 f)

$8 \times 2^x + 3 \times 2^1 \times 2^x = 14 \times 2^x \text{ or } 7 \times 2^{x+1}$

13 g)

$2^4 \times 2^{2x} + 2^{2x} = 17 \times 2^{2x}$

14 a)

$\frac{x^3+1}{x^4y+x}$

14 b)

$\frac{x^4+xy}{x^3y^3+y}$

14 c)

$\frac{x^6z-yz^4}{x^4z+x^2y^{\frac{1}{2}}}$

14 d)

$\frac{x^8y^{\frac{1}{2}}-x^2y^{\frac{3}{2}}z^4}{y^{\frac{1}{2}}+x^3z^2}$

15 a)

$x = \frac{1}{2}$

15 b)

$x = 1$

15 c)

$3^{2x} \times 3^{2x} = 3^2$

$x = \frac{1}{2}$

15 d)

$2^x = 2^{4+\frac{1}{2}x}$

$x = 4 + \frac{1}{2}x \Rightarrow x = 8$

16)

$$\begin{aligned}(2^2)^{x+2} - 2^{2x+3} &= 2^4\\ 2^{2x+4} - 2^{2x+3} &= 2^4\\ 2^4 \times 2^{2x} - 2^3 \times 2^{2x} &= 2^4\\ 8 \times 2^{2x} &= 2^4\\ 2^3 \times 2^{2x} &= 2^4\\ 2x + 3 = 4 \Rightarrow x &= \frac{1}{2}\end{aligned}$$

17)

$$\begin{aligned}2^{501} - 2^{497} &= 30 \times 64^k\\ 2^{497} \times (2^4 - 1) &= 30 \times (2^6)^k\\ 2^{497} \times 15 &= 30 \times (2^6)^k\\ 2^{497} &= \frac{30}{15} \times (2^6)^k\\ 2^{497} &= (2^1) \times (2^6)^k\\ 497 = 1 + 6k \Rightarrow k &= \frac{248}{3}\end{aligned}$$

18)

$$\begin{aligned}\frac{2^x}{2^5 \times 2^{\sqrt{x}}} = 2 \Rightarrow 2^x &= 2^1 \times 2^5 \times 2^{\sqrt{x}}\\ \Rightarrow x &= 6 + x^{\frac{1}{2}}\\ \Rightarrow x - x^{\frac{1}{2}} - 6 = 0 &\Rightarrow (x^{\frac{1}{2}} - 3)(x^{\frac{1}{2}} + 2) = 0\end{aligned}$$

$x^{\frac{1}{2}} = 3 \Rightarrow x = 9$

$x^{\frac{1}{2}} = -2$ cannot be a solution since $\sqrt{x} \geq 0$.

19)

$(x^2 - 4x - 1)^{3x+2} = 4^{-1}$

$x^2 - 4x - 1 = 4 \Rightarrow x = 5,\ x = -1$

$3x + 2 = -1 \Rightarrow x = -1$

$\therefore x = -1$

(or $x = -0.2988475575\ldots$)